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Module 3 – Databases: Data Limitations Reflection

Paragraph 1 – Bias and Self-Selection

Grad Cafe data is entirely self-reported by applicants who chose to submit their information anonymously. This creates significant selection bias: applicants who posts tend to be more engaged, more anxious about outcomes, or more likely to attend elite programs – meaning the dataset does not represent the full population of graduate applicants. For example, the average GRE quantitative score in the national 2022-2023 data was 157, yet reported scores on Grad Cafe tend to skew higher (~165), suggesting that higher-performing applicants are disproportionately represented. Similarly, GPA and GRE fields were missing for nearly all of our 30,000 entries, indicating that most users either do not provide these metrics or that Grad Cafe does not surface them in the standard listing view. This missingness is not random – it is likely correlated with applicant confidence, program prestige, or outcome, making any averages we compute unreliable as population estimates.

Paragraph 2 – Reliability and Representativeness

Beyond selection bias, the data suffers from issues of reliability and standardization. Program names, university names, and statuses are entered freely by users with no validation, resulting in inconsistencies such as "JHU," "Johns Hopkins," and "John Hopkins" all referring to the same institution. Our LLM standardization step partially addressed this, but abbreviations, misspellings, and ambiguous entries remain. Additionally, the start term field (Fall/Spring/Year) was not captured in our scrape because Grad Cafe only surfaces it on individual entry pages, not the summary table – meaning any term-based analysis is limited. Together, these limitations mean that while Grad Cafe data is valuable for identifying broad trends in graduate admissions, it should not be treated as a representative or complete dataset. Any conclusions drawn – such as acceptance

rates or average GPAs — must be interpreted cautiously, with the understanding that the data reflects only a vocal, self-selected subset of the true applicant pool.